

The same pattern, five costumes

Fibonacci, the gadgets **L** **F** **C**, and the categorical trace **⊗** — all one model-checking invariant (TransitionSystem / Reachable / CIC)

Visual companion to RequestProject/ModelChecking/FibonacciTracePattern.lean (sorry-free)

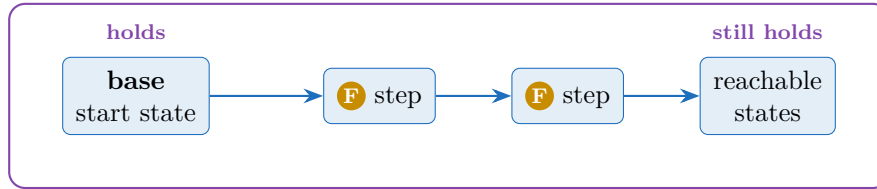
Your question. The picture books say Theorem 6 has a **Fibonacci pattern**, is built from the three gadgets **L** linearized trace, **F** Frobenius, **C** cyclotomic cosets, and from one **categorical gadget** **⊗** (the evaluation/coevaluation *trace loop*, “what returns”). *Are those same patterns already inside the model-checking core — base/step, Reachable, TransitionSystem, the cross invariant, and the CIC reading?*

Verified answer: yes. Each one is literally an instance of a single schema, and the accompanying Lean file proves it end to end, sorry-free:

base case holds + every step preserves it $\implies$ it holds at every reachable state.

That one theorem is ModelChecking.InductiveInvariant.reachable; and (see CICLink.lean) it *is* CIC induction — Lean’s auto-generated recursor Reachable.rec. Below, every ✓ box names a checked declaration.

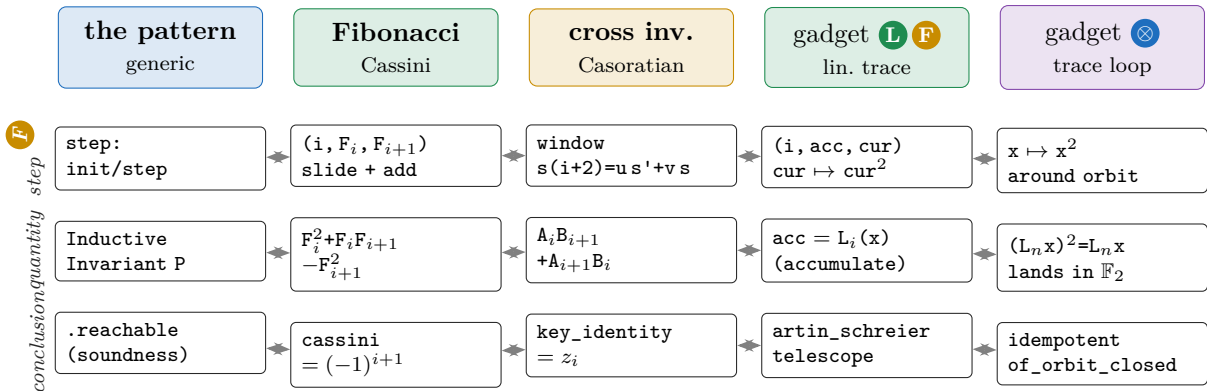
0. The shared skeleton (proved once)



invariant / conserved (or accumulated) quantity holds at every step

TransitionSystem = (init, step); Reachable = base + step (a CIC inductive); InductiveInvariant = (init_holds, step_preserves); soundness = InductiveInvariant.reachable.

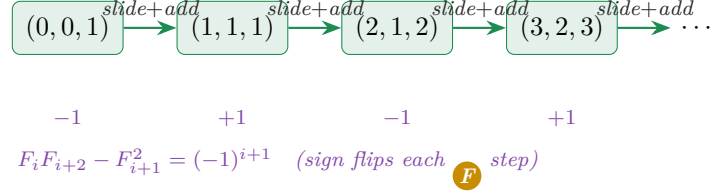
1. The five costumes, side by side



$\longleftrightarrow$ each grey link = “same shape, different words”. All boxes ✓ in FibonacciTracePattern.lean.

2. The Fibonacci pattern *is* a transition system; Cassini *is* a cross invariant

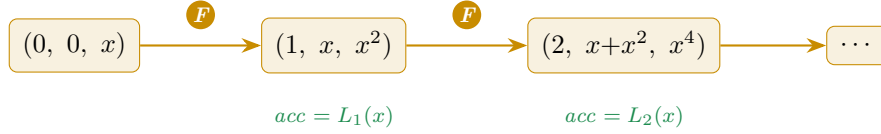
The two-step recurrence $F_{i+2} = F_{i+1} + F_i$ is a TransitionSystem on the window (i, F_i, F_{i+1}) . Cassini’s identity is the *sign-alternating* inductive invariant — the Fibonacci twin of Theorem 6’s Casoratian.



✓ `fibSystem, fibWindow_reachable, cassiniInv, cassini_is_inductiveInvariant, cassini`. Kin: `CrossInvariant.lean (key_identity_via_reachable), Theorem6.lean (casoratian, cross_invariant)`.

3. The gadgets $\mathbf{L}$ and $\mathbf{F}$: the linearized trace is a folded accumulator

Frobenius $\mathbf{F}$ ($x \mapsto x^2$) is the *step*. The linearized trace $\mathbf{L}$, $L_k(x) = \sum_{i < k} x^{2^i}$, is exactly the *accumulator* carried along the run — so $\mathbf{L}$ is “ $\mathbf{F}$ folded”.

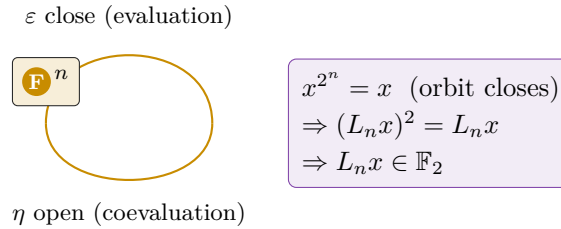


The $\mathbf{L}$ – $\mathbf{F}$ glue (equation (1) of the paper), an invariant of the run: $L_k(x^2 + x) = x^{2^k} + x$.

✓ `truncTrace, traceSystem, traceInv, trace_is_inductiveInvariant, traceWindow_reachable, truncTrace_artin_schreier`.

4. The categorical gadget $\otimes$: “what returns” = the loop closes

The trace loop opens a wire (coevaluation), lets $\mathbf{F}$ walk the orbit, and closes it (evaluation): what survives is the *round trip*. When the Frobenius orbit *closes* ($x^{2^n} = x$), the accumulated trace becomes a fixed point of $\mathbf{F}$ — it is idempotent, so it lands in the ground field $\mathbb{F}_2$: the string-diagram slogan “the closed loop returns to the unit object I ”.



✓ `truncTrace_frobenius` ($\mathbf{F}$ commutes with $\mathbf{L}$), `truncTrace_idempotent_of_orbit_closed` (loop closes $\Rightarrow$ lands in the base).

5. The gadget $\mathbf{C}$: the survivors of the loop form a subgroup

The fixed points of $\mathbf{F}^k$ (the *surviving loops* = cyclotomic cosets, gadget $\mathbf{C}$) are closed under addition (freshman’s dream): so “what returns” is an additive subgroup — exactly the coset bookkeeping $\mathbf{C}$ counts.

✓ `frob_fixed_closed_add`: $x^{2^k} = x, y^{2^k} = y \Rightarrow (x+y)^{2^k} = x+y$.

6. Master table

Paper pattern	Model-checking name	One shared idea	Lean (...Pattern.lean)
Fibonacci recursion	Transition System + Reachable	base + two-step slide generates the run	<code>fibSystem</code> , <code>fibWindow_reachable</code> ✓
Cassini identity	inductive invariant	quantity fixed (up to sign) by every step	<code>cassini_is_inductiveInvariant</code> , <code>cassini</code> ✓
cross invariant (Thm 6)	inductive invariant	Casoratian conserved: $= z_i$ at every i	<code>CrossInvariant.key_identity_via_reachable</code> ✓
F Frobenius	step	the next-state map $x \mapsto x^2$	<code>traceSystem.step</code> ✓
L linearized trace	accumulated invariant	acc folded along the F -run	<code>trace_is_inductiveInvariant</code> ✓
C cyclotomic cosets	the reachable/return set	fixed points of F ^k (a subgroup)	<code>frob_fixed_closed_add</code> ✓
⊗ categorical trace	Reachable closes = CIC recursor	“what returns”; closed loop lands in I	<code>truncTrace_idempotent_of_orbit_closed</code> ✓
soundness	Inductive-Invariant.-reachable	one theorem for all rows; it is <code>Reachable.rec</code>	<code>CICLink.Inductive.reachable'</code> ✓

7. Verdict

Are the Fibonacci pattern, the gadgets **L **F** **C**, and the categorical trace **⊗** all present in the model-checking / CIC core?** Yes — and it is now machine-checked. Each is the single move $base + step \Rightarrow reachable$: **F** is the **step**; **L** is the accumulator folded along the run; Fibonacci/Cassini and the Theorem-6 cross invariant are the conserved quantity; **C** is the return/fixed-point set; and **⊗**, “what returns”, is precisely the **Reachable** inductive closing on itself — whose soundness *is* the CIC recursor `Reachable.rec`. One theorem, `InductiveInvariant.reachable`, discharges all of them.

A lens, not a magic wand. The pattern unifies the *shape* of these devices and this file proves the shape end to end. It does not, by itself, deliver the paper’s genuinely extra mathematics — the Fibonacci *weight* recursion $w(A_{i+2}) = w(A_{i+1}) + w(A_i)$ with $w(R) = 2F_{k'} - 1$, or the Gauß/Jacobi-sum evaluation behind the difference sets. Those still need their own arguments; the pattern says *what to look at*, uniformly: base, step, and what is conserved or returns.